25900517

MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) Regular EXAMINATION, October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCMBG200 - GENERAL MICROBIOLOGY
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions: Answer **any 10** questions
Each question carries **2 marks**

1. **Give a short note on bacterial G+C content.** **[U] / [CO1]**
   Ans: Explain G+C content and its role in classification (1), calculation (1)
2. **Mention the function of the chromophore group and auxochrome group.** **[K] / [CO2]**
   Ans: Chromophore – imparts colouring property (1), auxochrome - ionic charge allows the stain to bind to the oppositely charged parts of the cell (1)
3. **What are the applications of dry heat sterilisation?** **[K] / [CO3]**
   Ans: Inoculation loops, tips of forceps, scalpels: sterilised by quickly passing them through a flame - flaming. Glassware: petri dishes, slides, pipettes, and test tubes. Metal instruments: such as scissors, forceps, scalpels and needles. Paraffin oil, powders – hot air oven.
4. **Name any one anaerobic culture media.** **[K] / [CO2]**
   Ans: Robertson’s cooked meat medium / Thioglycollate medium (Any one two marks)
5. **What is Bergey's manual?** **[K] / [CO1]**
   Ans: Explain Bergey's Manual – David H Bergey - for the classification and identification of bacteria and other prokaryotes. It is divided into two main versions: Bergey's Manual of Determinative Bacteriology, used for identifying unknown cultures based on phenotypic characteristics, and Bergey's Manual of Systematic Bacteriology, which classifies organisms based on evolutionary relationships (phylogeny) using genetic data (2).
6. **Mention two functions of calcium-dipicolinic acid in endospores.** **[K] / [CO1]**
   Ans: Confers resistance to heat and other stresses, primarily by stabilising and protecting the DNA. It works by binding water, which removes it from the core, and by forming a complex with calcium ions that helps stabilise the DNA against damage (2).

7. **Name two Gram-positive cocci.** **[K] / [CO2]**

   Ans: Staphylococci / Streptococci / Micrococci / Sarcina (Any two – 1 mark each)

8. **Mention the standard chemical used in the test for comparing disinfectants. [K] / [CO3]**

   Ans: Phenol (2)

9. **Define extracellular polymeric substance (EPS) in bacteria.** **[K] / [CO1]**

   Ans: Explain capsule and slime layer - Many bacteria secrete a viscid material around the cell surface. When this is organised into a sharply defined structure, it is known as the capsule; when it is a loose, undemarcated secretion, it is called the slime layer (2).

10. **Mention one advantage of the spread plate method.** **[U] / [CO2]**

    Ans: Both a quantitative method for counting microorganisms and a qualitative method for isolating support colony morphology study, highly sensitive, quick, cost-effective and easy method – Explain any one advantage 2 marks.

11. **What is the periplasmic space?** **[U] / [CO1]**

    Ans: Space between the cytoplasmic membrane and the bacterial outer membrane in Gram-negative bacteria (2)

12. **Give a note on the bacterial chromosome.** **[U] / [CO1]**

    Ans: Explain bacterial chromosome - A single, circular, double-stranded DNA molecule, highly compacted/condensed and located in a region of the cytoplasm called the nucleoid (2).

[2x10 = 20]

Part B

Short Essay Questions: Answer **any 4** questions
Each question carries **5 marks**

13. **Describe the mechanism of microbial destruction by UV radiation.** **[K] / [CO3]**

Ans: Explain lethal role of UV radiation – UV light energy is absorbed by nucleic acids - formation of pyrimidine dimers - inhibits or kills bacteria by damaging their DNA, due to the formation of lesions (mutations) that stop replication and reproduction (3), Application of UV radiation – sterilizing enclosed areas such as biosafety cabinets in laboratories, entryways, Operation theatres and laboratories (2).

14. **Explain the fluid-mosaic model of the bacterial cell membrane.** **[U] / [CO1]**

Ans: Explain cytoplasmic membrane - semipermeable, structure and composition - lipid bilayer, amphipathic nature, external and internal proteins, hopanoids (5)

15. Discuss non-beta-lactam antibiotics. **[U] / [CO3]**

Ans: Explain the mode of action and uses of Bacitracin (2.5) and Cycloserine (2.5)

16. Differentiate between pili and fimbriae in terms of structure and function. [U] / [CO1]

Ans: Explain Fimbriae and pili – structure (2), types and function (3)

17. Describe the principle and procedure of endospore staining. **[K] / [CO2]**

Ans: Endospore staining - Principle – Heat method - Malachite green (primary stain), decolouriser – water and counter stain - safranine (2.5). Procedure (2.5)

18. Discuss the factors that influence sterilisation by heat. **[U] / [CO3]**

Ans: Explain factors – nature of heat, temperature and time, number and characteristics of organisms and material from which it is eradicated (5)

[5x4 = 20]

Part C

Long Essay Questions Answer **any 1** question
Each question carries **10 marks**

19. Write an essay on the composition, structural organisation and function of peptidoglycan. **[U] / [C01]**

Ans: Peptidoglycan – Composition – NAM, NAG (β – β-1,4-glycosidic bond), amino acids – L alanine, D glutamic acid, L lysine, D alanine – pentaglycine peptide bond/crosslinking (4), Structure - Diagram (4). Function – Rigidity and structural integrity, Protection against osmotic lysis (2)

20. Discuss the applications of lawn culture. Explain the stab and stroke culture.[U] / CO2]

Ans: Explain the method and application of lawn culture (4). Explain the method & application of stab culture (3) and Stroke culture (3)

[10x1 = 10]